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A thermodynamic description of the Gd–Mg–Sm system

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ABSTRACT

The Mg–Sm, Gd–Sm and Gd–Mg–Sm systems were thermodynamically optimized using the CALPHAD technique. The solution phases, liquid, bcc, hcp and rhombohedral, were described by the substitutional solution model. The isostructural compounds, MgGd in the Gd–Mg system and MgSm in the Mg–Sm system with a B2 structure was assumed to form a continuous range of solid solutions in the Gd–Mg–Sm system. The order–disorder transition between the bcc solution with an A2 structure and compound Mg(Gd, Sm) with a B2 structure in the system has been taken into account and thermodynamically modeled. The other isostructural compounds Mg_5Gd and Mg_5Sm, Mg_3Gd and Mg_3Sm, Mg_2Gd and Mg_2Sm in the Gd–Mg–Sm system were described according to the formulae Mg_5(Gd,Sm), Mg_3(Gd,Sm), and Mg_2(Gd,Sm), respectively. The compound $Mg_{41}Sm_5$ with a homogeneity range was treated as a line compound $Mg_{41}(Gd, Sm)_5$ in the Gd–Mg–Sm system. Based on the experimental data in the Mg-rich corner of the Gd–Mg–Sm system, a set of thermodynamic parameters describing the Gibbs energies of individual phases of the Gd–Mg–Sm system as functions of composition and temperature was obtained. In addition, the complete ternary phase diagram of the Gd–Mg–Sm system were predicted.

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1. Introduction

Mg–Gd–Sm alloys are interesting because both Gd and Sm are quite good prospects for improving the strength properties of magnesium alloys [1,2]. In traditional alloy development, experimental investigations with many different alloy compositions are performed. The selection criteria for multi-component alloying elements and their compositions become diffused in a traditional approach. Computational thermochemistry as used in the CALPHAD approach can provide a clear guideline for such selections and help to avoid large scale experiments with less promising alloys. Thus, it is a powerful tool to cut down on cost and time during the development of Mg-alloys.

As a part of the Mg-Alloys database [3], in the present work, thermodynamic modeling and optimization of the Mg–Sm, Gd–Sm and Gd–Mg–Sm systems are performed using the CALPHAD method on the basis of the experimental data reported in the literature.

2. Experimental information

2.1. Gd–Mg system

Based on the experimental data of Joseph and Gschneidner [4], Rokhlin and Nikitina [5] and Manfrinetti and Gschneidner [6], Cacciamani et al. [7,8] and Guo et al. [9] optimized the Gd–Mg system. In the present work, the thermodynamic parameters of Guo et al. [9] were adopted and the calculated phase diagram was shown in Fig. 1.

2.2. Gd–Sm system

Lundin and Yamamoto [10] determined the liquidus, solidus, the bcc–hcp transformation temperatures and the homogeneity range of the rhombohedral phase in the Gd–Sm system. Torchinova et al. [11] also studied the Gd–Sm phase diagram. The partial phase diagram that they published is in reasonable accord with the diagram presented by Lundin and Yamamoto [10], except the homogeneity range of the rhombohedral phase. Torchinova et al. [11] suggested that the rhombohedral phase exists from 41.9–100 at.% Sm, which is a wider range than that of 60–100 at.% Sm reported by Lundin and Yamamoto [10]. Beaudry and Gschneidner [12] modified the melting and transition temperatures of Gd and Sm elements.

Based on the above experimental data, Gschneidner and Calderwood [13] constructed and assessed the Gd–Sm system. Four solution phases, liquid, bcc, hcp and rhombohedral were included, and the homogeneity range of the rhombohedral phase determined by Lundin and Yamamoto [10] was adopted in their work.

Lundin and Yamamoto [10] determined the thermodynamic activities of Sm in the hcp phase in the temperature range 1073–1373 K. Using the activity data of Sm in the hcp phase, the activity of Gd, the integral free energies, enthalpies and entropies of mixing at 1173 K were calculated.

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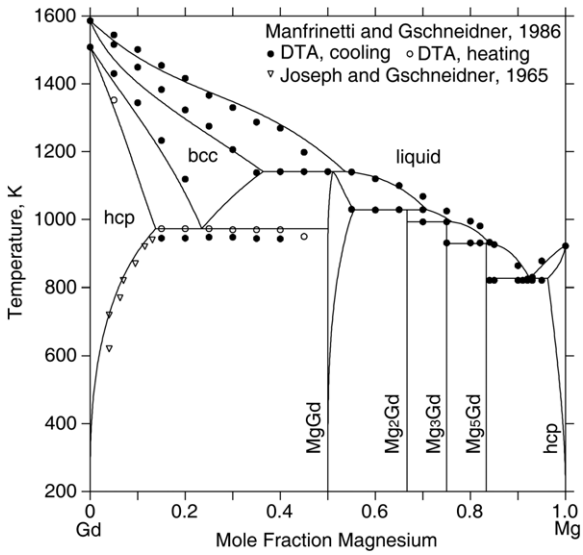

Fig. 1. Calculated Gd–Mg phase diagram using the thermodynamic description determined by Guo et al. [9] and comparison with the experimental data [4,6].

Kohji et al. [14] measured the magnetic properties at 4.2 K and paramagnetic susceptibility up to room temperature on the Gd–Sm single crystals and polycrystals less than 30 at.% Sm concentration. Fujimori et al. [15] measured the magnetic properties of some Gd–Sm and Gd–Nd alloys. Arajs et al. [16] determined the ferromagnetic Curie temperatures Tc of Gd alloys containing 5, 10, 20, 25, 30, and 35 at.% Sm. D'Onofrio et al. [17] investigated the effect of 5% of Pr, Nd, Gd and Tb on the magnetic ordering temperatures of Sm by susceptibility and resistivity measurements. The Magnetic properties determined by the above researchers [15–17] are below 298 K, the Curie temperature and Bohr magneton in the Gd–Sm system are not considered in the present work.

2.3. Mg–Sm system

On the basis of the experimental data of Saccone et al. [18], Jia et al. [19] optimized the Mg–Sm system thermodynamically, in which the order-disordered transition between bcc(cI2-W(A2)) and MgSm(cP2-CsCl(B2)) phases was not considered. In the present work, the thermodynamic description of the Mg–Sm system was modified.

2.4. Gd–Mg–Sm system

Rokhlin et al. [20] investigated phase equilibria in the Mg-rich corner at 573 and 773 K using optical and scanning electron microscopy, X-ray diffraction and electrical resistivity measurements. Two isothermal sections at 573 and 773 K in the Mg-rich corner were constructed. Rokhlin et al. [21] furthermore reported four vertical sections at 90.8 at.% Mg, 94.5 at.% Mg, 3 at.% Sm, and $x(\mathrm{Sm})/x(\mathrm{Gd}) = 3$. In addition, Rokhlin et al. [21] also constructed the liquidus surface in the Mg-rich corner of this system. Based on the above experimental data, Ferro et al. [22] reviewed the Gd–Mg–Sm system.

3. Thermodynamic models

In order to maintain consistency with other thermodynamic assessments, the Gibbs energy of each pure element is taken from the SGTE (Scientific Group Thermodata Europe) pure elements database compiled by Dinsdale [23]. The Gibbs energy function is described by the following equation:

$$G_i^{\phi}(T) = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9}. \quad (1)$$

The Gibbs energy of the element i, $G_i^{\phi}(T)$, in a standard element reference (SER) state, is denoted by GHSER_i, i.e.,

$$\mathrm{GHSER}_i = {}^0G_i^{\phi}(T) - H_i^{\mathrm{SER}}(298.15\ \mathrm{K}) \quad (2)$$

where $H_i^{\mathrm{SER}}(298.15\ \mathrm{K})$ is the molar enthalpy of the element i at 298.15 K in its SER state, hcp for Gd and Mg, and rhombohedral for Sm.

3.1. Solution phases

In the Gd–Mg–Sm system, there are four solution phases: liquid, bcc, hcp and rhombohedral. Their molar Gibbs energies are described by the following expression:

$$\begin{aligned} G_{\mathrm{m}}^{\phi} &= x_{\mathrm{Gd}}G_{\mathrm{Gd}}^{\phi}(T) + x_{\mathrm{Mg}}G_{\mathrm{Mg}}^{\phi} + x_{\mathrm{Sm}}G_{\mathrm{Sm}}^{\phi}(T) \\ &\quad + RT\left(x_{\mathrm{Gd}}\ln x_{\mathrm{Gd}} + x_{\mathrm{Mg}}\ln x_{\mathrm{Mg}} + x_{\mathrm{Sm}}\ln x_{\mathrm{Sm}}\right) \\ &\quad + {}^{E}G_{\mathrm{m}}^{\phi} + {}^{\mathrm{mg}}G_{\mathrm{m}}^{\phi} \end{aligned} \quad (3)$$

where R is the gas constant; x_{Gd}, x_{Mg} and x_{Sm} are the mole fraction of Gd, Mg and Sm, respectively; ${}^{E}G_{\mathrm{m}}^{\phi}$, which is the excess Gibbs energy, and is expressed by the Redlich–Kister polynomial,

$$\begin{aligned} {}^{E}G_{\mathrm{m}}^{\phi} &= x_{\mathrm{Gd}}x_{\mathrm{Mg}}\sum_j {}^{j}L_{\mathrm{Gd,Mg}}^{\phi}(x_{\mathrm{Gd}} - x_{\mathrm{Mg}})^j \\ &\quad + x_{\mathrm{Gd}}x_{\mathrm{Sm}}\sum_j {}^{j}L_{\mathrm{Gd,Sm}}^{\phi}(x_{\mathrm{Gd}} - x_{\mathrm{Sm}})^j + x_{\mathrm{Mg}}x_{\mathrm{Sm}} \\ &\quad \times \sum_j {}^{j}L_{\mathrm{Mg,Sm}}^{\phi}(x_{\mathrm{Mg}} - x_{\mathrm{Sm}})^j + x_{\mathrm{Gd}}x_{\mathrm{Mg}}x_{\mathrm{Sm}}{}^{j}L_{\mathrm{Gd,Mg,Sm}}^{\phi} \end{aligned} \quad (4)$$

where ${}^{j}L_{\mathrm{Gd,Mg}}^{\phi}$, ${}^{j}L_{\mathrm{Gd,Sm}}^{\phi}$ and ${}^{j}L_{\mathrm{Mg,Sm}}^{\phi}$ are the interaction parameters between elements Gd and Mg, Gd and Sm, and Mg and Sm, respectively; ${}^{j}L_{\mathrm{Gd,Mg,Sm}}^{\phi}$ is the ternary interaction parameter; ${}^{j}L_{\mathrm{Gd,Sm}}^{\phi}$, ${}^{j}L_{\mathrm{Mg,Sm}}^{\phi}$ and ${}^{j}L_{\mathrm{Gd,Mg,Sm}}^{\phi}$ are to be evaluated in the present work. The general form of the interaction parameters is

$$L^{\phi} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1}. \quad (5)$$

In most cases, only the first one or two terms are used according to the temperature dependence on the experimental data. ${}^{\mathrm{mg}}G_{\mathrm{m}}^{\phi}$ is the magnetic contribution to the Gibbs energy, and ${}^{\mathrm{mg}}G_{\mathrm{m}}^{\mathrm{hcp}}$ the magnetic contribution to the Gibbs energy in the hcp phase of the Gd–Sm system. Due to a lack of experimental data about magnetism, the interaction parameters in ${}^{\mathrm{mg}}G_{\mathrm{m}}^{\phi}$ were not optimized in the present work.

3.2. Intermetallic compounds

3.2.1. Line intermetallic compounds

The intermetallic compounds Mg_5Gd and Mg_5Sm, Mg_3Gd and Mg_3Sm, Mg_2Gd and Mg_2Sm in the binary Gd–Mg and Mg–Sm systems have the same crystal structure and are treated as line compounds $Mg_5(\mathrm{Gd, Sm})$, $Mg_3(\mathrm{Gd, Sm})$, and $Mg_2(\mathrm{Gd, Sm})$ in the Gd–Mg–Sm system. The compound $Mg_{41}Sm_5$ with a homogeneity range was treated as a line compound $Mg_{41}(\mathrm{Gd, Sm})_5$ in the Gd–Mg–Sm system. The Gibbs energy per mole of the formula unit $Mg_m(\mathrm{Gd, Sm})_n$, $G_{\mathrm{m}}^{\mathrm{Mg}_m\mathrm{R}_n}$, is expressed as follows:

$$\begin{aligned} G_{\mathrm{m}}^{\mathrm{Mg}_m\mathrm{R}_n} &= y_{\mathrm{Gd}}G_{\mathrm{Mg:Gd}}^{\mathrm{Mg}_m\mathrm{R}_n} + y_{\mathrm{Sm}}G_{\mathrm{Mg:Sm}}^{\mathrm{Mg}_m\mathrm{R}_n} + nRT(y_{\mathrm{Gd}}\ln y_{\mathrm{Gd}} + y_{\mathrm{Sm}}\ln y_{\mathrm{Sm}}) \\ &\quad + y_{\mathrm{Gd}}y_{\mathrm{Sm}}\sum_j {}^{j}L_{\mathrm{Mg:Gd,Sm}}^{\mathrm{Mg}_m\mathrm{R}_n}(y_{\mathrm{Gd}} - y_{\mathrm{Sm}})^j \end{aligned} \quad (6)$$

Table 1
Thermodynamic parameters in the Gd–Mg–Sm system[a].

Phase	Thermodynamic parameters		Reference
liquid:(Gd, Mg, Sm)$_1$	G(liquid, Gd) =		[23]
	298–1000	$+13060.0262 - 8.32179974T + \mathrm{GHSER_{Gd}}$	
	1000–1508	$+146262.037 - 1208.70685T + 159.352082T \ln T$	
		$- 0.108247135T^2 + 1.06945505 \times 10^{-5}T^3 - 19678357T^{-1}$	
	1508–3600	$-5397.314 + 192.336215T - 38.5075T \ln T$	
	G(liquid, Mg) =		[23]
	298–923	$+8202.243 - 8.83693T + \mathrm{GHSER_{Mg}} - 8.0176 \times 10^{-20}T^7$	
	923–3000	$-5439.869 + 195.324057T - 34.3088T \ln T$	
	G(liquid, Sm) =		[23]
	298–1190	$+3468.783 + 20.117456T - 11.6968284T \ln T$	
		$- 0.032418177T^2 - 4.544272 \times 10^{-6}T^3 + 23528T^{-1}$	
	1190–2100	$-11728.229 + 273.487076T - 50.208T \ln T$	
	${}^0L^{\mathrm{liq.}}_{\mathrm{Gd,Mg}} = -36681.3 + 16.2484T$		[9]
	${}^1L^{\mathrm{liq.}}_{\mathrm{Gd,Mg}} = +34233.8 - 10.7783T$		[9]
	${}^2L^{\mathrm{liq.}}_{\mathrm{Gd,Mg}} = -7352.9$		[9]
	${}^0L^{\mathrm{liq.}}_{\mathrm{Gd,Sm}} = -24873.8 - 9.4838T$		This work
	${}^1L^{\mathrm{liq.}}_{\mathrm{Gd,Sm}} = -6157.0$		This work
	${}^0L^{\mathrm{liq.}}_{\mathrm{Mg,Sm}} = -21155.4 - 0.4056T$		This work
	${}^1L^{\mathrm{liq.}}_{\mathrm{Mg,Sm}} = -37422.6 + 14.7798T$		This work
	${}^0L^{\mathrm{liq.}}_{\mathrm{Gd,Mg,Sm}} = +2457.9$		This work
	${}^1L^{\mathrm{liq.}}_{\mathrm{Gd,Mg,Sm}} = +144071.0$		This work
	${}^2L^{\mathrm{liq.}}_{\mathrm{Gd,Mg,Sm}} = 0.0$		This work
	$\mathrm{GHSER_{Gd}}$ =		[23]
	200–1000	$-6834.5855 + 97.13101T - 24.7214131T \ln T$	
		$- 0.00285240521T^2 - 3.14674076 \times 10^{-7}T^3$	
		$- 8665.73348T^{-1}$	
	1000–1508	$-6483.25362 + 95.6919924T - 24.6598297T \ln T$	
		$- 0.00185225011T^2 - 6.61211607 \times 10^{-7}T^3$	[23]
	1508–3600	$-123124.992 + 699.125537T - 101.800197T \ln T$	
		$- 0.0150644246T^2 + 6.39165948 \times 10^{-7}T^3$	
		$+ 29356890.3T^{-1}$	
	$\mathrm{GHSER_{Mg}}$ =		[23]
	298–923	$-8367.34 + 143.675547T - 26.1849782T \ln T$	
		$+ 4.858 \times 10^{-4}T^2 - 1.393669 \times 10^{-6}T^3 + 78950T^{-1}$	
	923–3000	$-14130.185 + 204.716215T - 34.3088T \ln T$	
		$+ 1.038192 \times 10^{28}T^{-9}$	
	$\mathrm{GHSER_{Sm}}$ =		[23]
	298–700	$-3872.013 - 32.10748T - 1.6485T \ln T$	
		$- 0.050254T^2 + 1.010345 \times 10^{-5}T^3 - 82168T^{-1}$	
	700–1190	$-50078.215 + 627.869894T - 102.665T \ln T$	
		$+ 0.0474522T^2 - 7.538383 \times 10^{-6}T^3 + 3861770T^{-1}$	
	1190–1345	$+289719.819 - 2744.50976T + 381.41982T \ln T$	
		$- 0.254986338T^2 + 2.7512152 \times 10^{-5}T^3 - 40102102T^{-1}$	
	1345–2100	$-23056.079 + 282.194375T - 50.208T \ln T$	
bcc (disordered part of B2): (Gd, Mg, Sm)$_1$(Va)$_3$	G(bcc, Gd : Va; 0) =		[23]
	200–1000	$+3233.80866 - 2.11184585T$	
		$+ 1.00000001 \times 10^{-7}T \ln T + \mathrm{GHSER_{Gd}}$	
	1000–1508	$+152792.743 - 1349.58873T + 180.097094T \ln T$	
		$- 0.119550229T^2 + 1.17915728 \times 10^{-5}T^3 - 22038836T^{-1}$	
	1508–1586	$-15783.7618 + 202.222057T - 38.960425T \ln T$	
	1586–3600	$-19850.5562 + 224.817909T - 41.904333T \ln T$	
		$+ 8.58222759 \times 10^{-4}T^2 - 3.77570269 \times 10^{-8}T^3$	
		$+ 995428.573T^{-1}$	
	G(bcc, Mg : Va; 0) =		[23]
	298–3000	$+3100 - 2.1T + \mathrm{GHSER_{Mg}}$	
	G(bcc, Sm : Va; 0) =		[23]
	298–1190	$-4368.72 + 55.972523T - 16.9298494T \ln T$	
		$- 0.025446016T^2 + 3.579527 \times 10^{-6}T^3 + 94209T^{-1}$	
	1000–1795	$-1297.79829 + 134.528352T - 27.3038477T \ln T$	
		$- 5.41757644 \times 10^{-4}T^2 - 3.05012175 \times 10^{-7}T^3$	
	1190–1345	$-15957.862 + 253.121044T - 46.9445T \ln T$	
	1345–2100	$+111191.653 - 624.680805T + 71.6856914T \ln T$	
		$- 0.047314968T^2 + 3.329865 \times 10^{-6}T^3 - 24870276T^{-1}$	

Table 1 (*continued*)

Phase	Thermodynamic parameters		Reference
	${}^{0}L^{bcc}_{Gd,Sm} = -24873.8 - 9.4838T$		This work
	${}^{1}L^{bcc}_{Gd,Sm} = -6157.0$		This work
	${}^{0}L^{bcc}_{Mg,Sm} = -40243.2 + 20.4276T$		This work
	${}^{1}L^{bcc}_{Mg,Sm} = -32374.2 + 18.7818T$		This work
	${}^{0}L^{bcc.}_{Gd,Mg} = -45347.5 + 25.5692T$		[9]
	${}^{1}L^{bcc.}_{Gd,Mg} = +10195.0 + 1.3355T$		[9]
	${}^{2}L^{bcc.}_{Gd,Mg} = +3267.1 - 3.5551T$		[9]
	${}^{0}L^{bcc}_{Gd,Mg,Sm} = 0.0$		This work
	${}^{1}L^{bcc}_{Gd,Mg,Sm} = +10000.0$		This work
	${}^{2}L^{bcc}_{Gd,Mg,Sm} = 0.0$		This work
hcp:(Gd, Mg, Sm)$_1$(Va)$_{0.5}$	$G(\text{hcp, Gd : Va; 0}) = +\text{GHSER}_{Gd}$		[23]
	$G(\text{hcp, Mg : Va; 0}) = +\text{GHSER}_{Mg}$		[23]
	$G(\text{hcp, Sm : Va; 0}) =$		
	298–700	$-3802.0356 - 32.176971T - 1.6485T\ln T - 0.050254T^2 + 1.010345\times10^{-5}T^3 - 82168T^{-1}$	
	700–1190	$-50008.2376 + 627.800403T - 102.665T\ln T + 0.0474522T^2 - 7.538383\times10^{-6}T^3 + 3861770T^{-1}$	
	1190–1345	$+289789.796 - 2744.57926T + 381.41982T\ln T - 0.254986338T^2 + 2.7512152\times10^{-5}T^3 - 40102102T^{-1}$	
	1345–2100	$-22986.1016 + 282.124884T - 50.208T\ln T$	
	${}^{0}L^{hcp}_{Gd,Mg} = -33346.6 + 19.3451T$		[9]
	${}^{1}L^{hcp}_{Gd,Mg} = +13854.0$		[9]
	${}^{0}L^{hcp}_{Gd,Sm} = -24873.8 - 9.4838T$		This work
	${}^{1}L^{hcp}_{Gd,Sm} = -6157.0$		This work
	${}^{0}L^{hcp}_{Mg,Sm} = +11086.3 - 19.5684T$		This work
	${}^{1}L^{hcp}_{Mg,Sm} = -9255.9$		This work
rhomb_C19:(Gd, Mg, Sm)$_1$	$G(\text{rhomb_C19, Gd; 0}) = +\text{GHSER}_{Gd} + 5000.0$		This work
	$G(\text{rhomb_C19, Mg; 0}) = +\text{GHSER}_{Mg} + 5000.0$		This work
	$G(\text{rhomb_C19, Sm; 0}) = +\text{GHSER}_{Sm}$		[23]
	${}^{0}L^{rhomb}_{Gd,Sm} = -38010.1 + 4.7246T$		This work
	${}^{1}L^{rhomb}_{Gd,Sm} = +2268.8$		This work
MgR (oedered part of B2):	$G^{MgR}_{Mg:Gd} = -28695.4 + 6.1567T$		[9]
(Gd, Mg, Sm)$_{0.5}$(Gd, Mg, Sm)$_{0.5}$	$G^{MgR}_{Mg:Gd} = -28695.4 + 6.1567T$		[9]
	$G^{MgR}_{Mg:Sm} = -41204.6 + 0.5539T$		This work
	$G^{MgR}_{Sm:Mg} = -41204.6 + 0.5539T$		This work
	${}^{0}L^{MgR}_{Gd,Mg:Gd} = -15446.6$		[9]
	${}^{1}L^{MgR}_{Gd,Mg:Gd} = -16247.3$		[9]
	${}^{0}L^{MgR}_{Gd:Mg,Gd} = -15446.6$		[9]
	${}^{1}L^{MgR}_{Gd:Mg,Gd} = -16247.3$		[9]
	${}^{0}L^{MgR}_{Mg:Mg,Sm} = {}^{0}L^{MgR}_{Mg,Sm:Mg} = -61420.5 + 8.0402T$		This work
	${}^{1}L^{MgR}_{Mg:Mg,Sm} = {}^{1}L^{MgR}_{Mg,Sm:Mg} = -19534.8$		This work
	${}^{0}L^{MgR}_{Mg:Gd,Sm} = {}^{0}L^{MgR}_{Gd,Sm:Mg} = +10000.0$		This work
	${}^{0}L^{MgR}_{Sm:Gd,Mg} = {}^{0}L^{MgR}_{Gd,Mg:Sm} = +15000.0$		This work
Mg_2R:Mg_2(Gd, Sm)	$G^{Mg_2R}_{Mg:Gd} = 2\text{GHSER}_{Mg} + \text{GHSER}_{Gd} - 50777.0 + 17.1490T$		[9]
	$G^{Mg_2R}_{Mg:Sm} = 2\text{GHSER}_{Mg} + \text{GHSER}_{Sm} - 35752.1 + 0.6912T$		This work
	${}^{0}L^{Mg_2R}_{Mg:Gd,Sm} = -13000.0$		This work

(*continued on next page*)

Table 1 (*continued*)

Phase	Thermodynamic parameters	Reference
Mg_3R:Mg_3(Gd, Sm)	$G^{Mg_3R}_{Mg:Gd} = 3GHSER_{Mg} + GHSER_{Gd} - 54289.8 + 13.0162T$	[9]
	$G^{Mg_3R}_{Mg:Sm} = 3GHSER_{Mg} + GHSER_{Sm} - 42053.8 + 0.4089T$	This work
	${}^0L^{Mg_3R}_{Mg:Gd,Sm} = -5000.0$	This work
Mg_5R : Mg_5(Gd, Sm)	$G^{Mg_5R}_{Mg:Gd} = 5GHSER_{Mg} + GHSER_{Gd} - 60521.6 + 11.2668T$	[9]
	$G^{Mg_5R}_{Mg:Sm} = 5GHSER_{Mg} + GHSER_{Sm} - 45811.4 + 1.7126T$	This work
$Mg_{41}R_5$:Mg_{41}(Gd, Sm)$_5$	$G^{Mg_{41}R_5}_{Mg:Gd} = 41GHSER_{Mg} + 5GHSER_{Gd} - 291122.4 + 46.3460T$	This work
	$G^{Mg_{41}R_5}_{Mg:Sm} = 41GHSER_{Mg} + 5GHSER_{Sm} - 232546.3 + 0.6032T$	This work

[a] In SI units (Joule, mole of the formula units and Kelvin).

where y_{Gd} and y_{Sm} are the site fractions of Gd and Sm on the second sublattice, respectively; the two parameters $G^{Mg_mR_n}_{Mg:Gd}$ and $G^{Mg_mR_n}_{Mg:Sm}$ represent the Gibbs energies of compounds Mg_mGd_n and Mg_mSm_n, referred to the pure elements in the SER state; the ${}^jL^{Mg_mR_n}_{Mg:Gd,Sm}$ represents the *j*th interaction parameter between elements Gd and Sm on the second sublattice.

3.2.2. Non-line intermetallic compounds

The compounds MgGd and MgSm have a B2 structure and form a continuous range of solid solution in the Gd–Mg–Sm ternary system, which is called MgR or Mg(Gd,Sm) in the following text, figures and tables. The order/disorder relation between bcc(cI2-W(A2)) and MgR(cP2-CsCl(B2)) phases has been taken into account and thermodynamically modeled. In order to indicate their structural relationship between the compound MgR (ordered, bcc-B2) and the bcc terminal solution phase (disordered, bcc-A2), in the present work, the two phases are modeled using a single sublattice formalism: $(Gd, Mg, Sm)_{0.5}(Gd, Mg, Sm)_{0.5}(Va)_3$. The Gibbs energy per mole of formula unit MgR with B2 structure is given by the following expression [24]:

$$G^{MgR}_m = G^{dis}_m(x_i) + G^{ord}_m(y'_i, y''_i) - G^{ord}_m(x_i) \quad (7)$$

where the mole fraction x_i and the site fractions y'_i, y''_i are related, according to $x_i = 0.5y'_i + 0.5y''_i$; the first term $G^{dis}_m(x_i)$ is the molar Gibbs energy of the disordered bcc phase, and the expression is the same as Eq. (3); $G^{ord}_m(y'_i, y''_i) - G^{ord}_m(x_i)$ is the ordering contributions to the Gibbs energy. When the phase becomes disordered, i.e. $y'_i = y''_i = x_i$, the contribution from the ordering, i.e. $G^{ord}_m(y'_i, y''_i) - G^{ord}_m(x_i)$, will be cancelled and only the first term $G^{ord}_m(x_i)$ gives a contribution to the Gibbs energy. $G^{ord}_m(y'_i, y''_i)$ is calculated using Eq. (10) in Ref [25], then $G^{ord}_m(x_i)$ is calculated using the same model parameters in Eq. (10) in Ref. [25] but replacing y'_i and y''_i with mole fraction x_i.

4. Assessment procedure

To obtain a thermodynamic description of a multi-component system, the thermodynamic description of each lower-order system is necessary.

The Gd–Mg binary system has been optimized by Cacciamani et al. [7,8], and Guo et al. [9]. In the present work, the thermodynamic parameters obtained by Guo et al. [9] were adopted.

The Mg–Sm binary system has been optimized by Jia et al. [19]. In the work of Jia et al. [19], the order-disordered transition between the bcc(cI2-W(A2)) and MgSm(cP2-CsCl(B2)) phases was not considered. Based on the thermodynamic parameters optimized by Jia et al. [19], the Mg–Sm system is re-optimized in the present work.

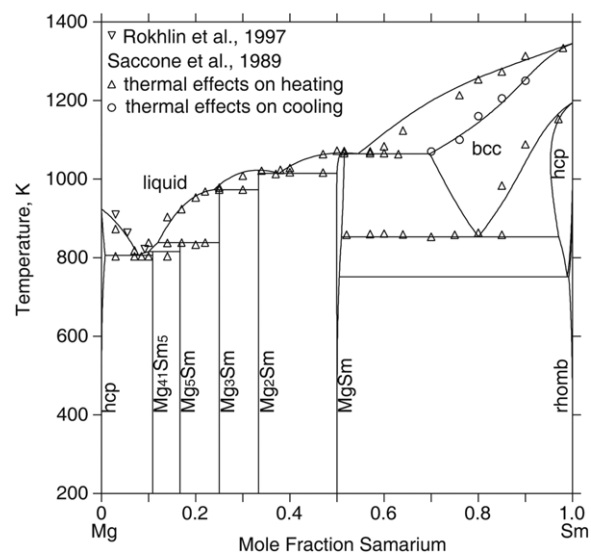

Fig. 2. Calculated phase diagram of Mg–Sm system using the present thermodynamic description and comparison with Saccone et al. [18] and Rokhlin et al. [21].

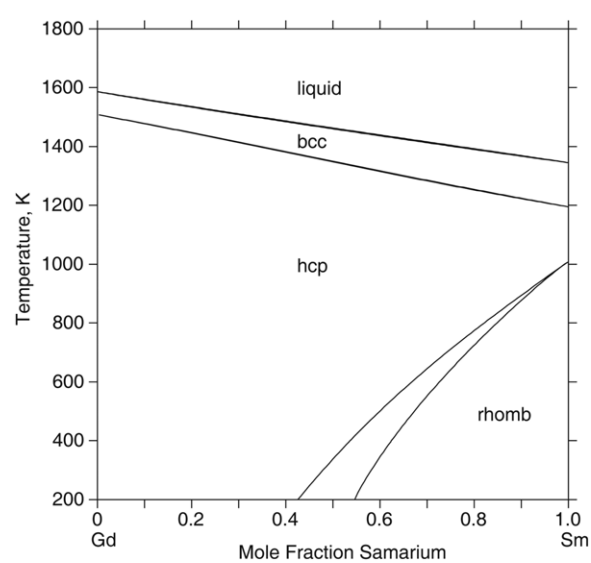

Fig. 3. Calculated phase diagram of Gd–Sm system using the present thermodynamic description.

In the Gd–Sm system, two elements Gd and Sm form four different solution phases, liquid, bcc, hcp, and rhombohedral at different temperature ranges. Only using the Gibbs energies functions of various phases offered by the PURE database [23], the

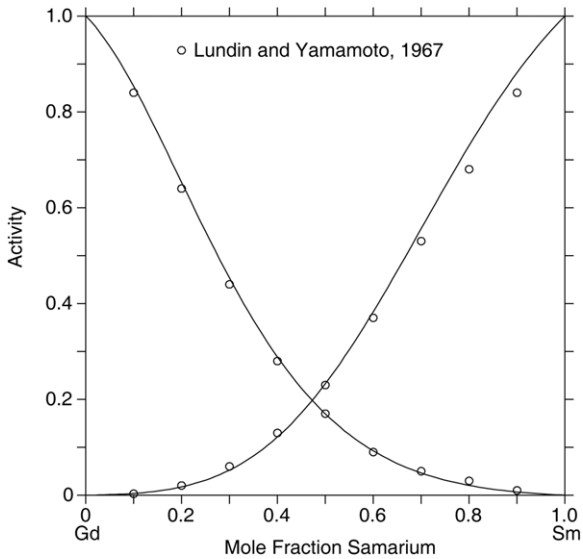

Fig. 4. Calculated activity of Gd and Sm in the hcp phase of Gd–Sm system at 1173 K and comparison with experimental data [10]. The reference states are hcp for Gd and Sm.

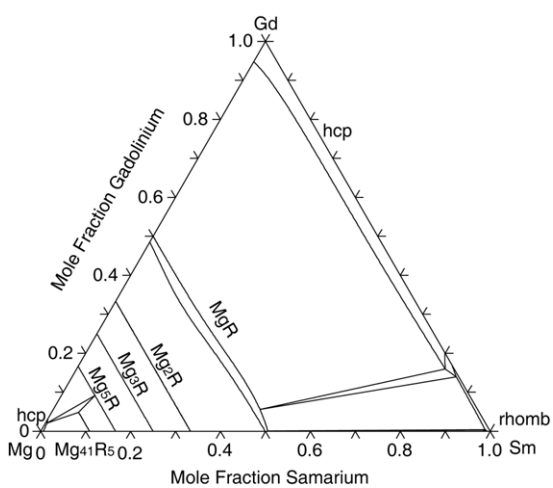

Fig. 5a. Calculated isothermal section of the Gd–Mg–Sm system at 773 K using the present thermodynamic description.

phase diagram of the Gd–Sm system [10] could be reproduced, except the phase equilibria between hcp and rhombohedral solutions. According to the thermodynamic properties of hcp and the phase equilibria between hcp and rhombohedral, the parameters ${}^{0}L^{\mathrm{hcp}}_{\mathrm{Gd,Sm}}$, ${}^{1}L^{\mathrm{hcp}}_{\mathrm{Gd,Sm}}$, ${}^{0}L^{\mathrm{rhomb}}_{\mathrm{Gd,Sm}}$ and ${}^{1}L^{\mathrm{rhomb}}_{\mathrm{Gd,Sm}}$ were optimized. Because of the addition of ${}^{0}L^{\mathrm{hcp}}_{\mathrm{Gd,Sm}}$ and ${}^{1}L^{\mathrm{hcp}}_{\mathrm{Gd,Sm}}$, the phase equilibria between hcp and bcc, bcc and liquid can not be reproduced using the Gibbs energy functions of bcc and liquid offered by the PURE database [23]. In order to reproduce the phase equilibria between hcp and bcc as well as bcc and liquid and simplify the thermodynamic parameters for the bcc and liquid phases, the following assumptions were made:

$$ {}^{0}L^{\mathrm{liq.}}_{\mathrm{Gd,Sm}} = {}^{0}L^{\mathrm{bcc}}_{\mathrm{Gd,Sm}} = {}^{0}L^{\mathrm{hcp}}_{\mathrm{Gd,Sm}} \quad (8) $$

$$ {}^{1}L^{\mathrm{liq.}}_{\mathrm{Gd,Sm}} = {}^{1}L^{\mathrm{bcc}}_{\mathrm{Gd,Sm}} = {}^{1}L^{\mathrm{hcp}}_{\mathrm{Gd,Sm}}. \quad (9) $$

Based on the above work of three binary systems and the experimental data of Rokhlin et al. [20,21], the Gd–Mg–Sm ternary system was optimized.

The thermodynamic description of the liquid, bcc, hcp and rhombohedral phases were obtained by a combination of the

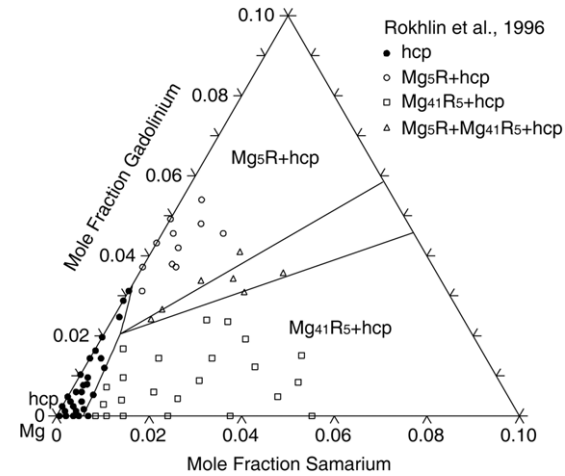

Fig. 5b. Enlarged section of Fig. 5a and comparison with experimental data of [20].

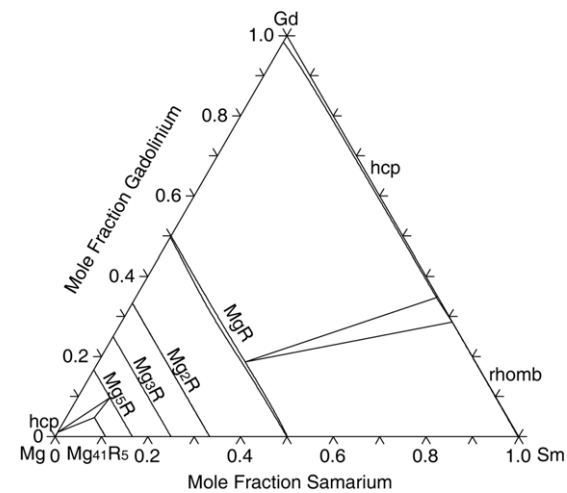

Fig. 6a. Calculated isothermal section of the Gd–Mg–Sm system at 573 K using the present thermodynamic description.

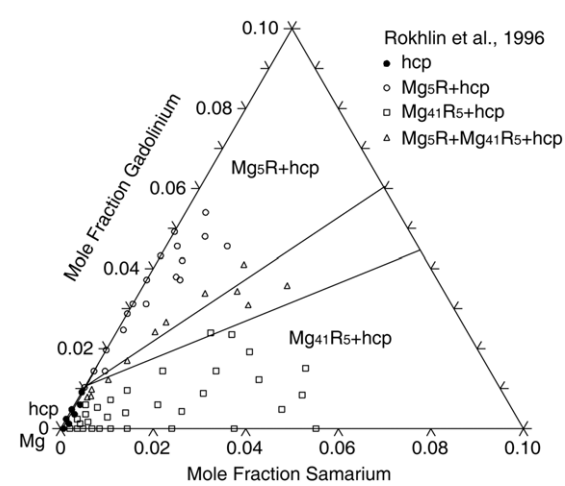

Fig. 6b. Enlarged section of Fig. 6a and comparison with experimental data of [20].

corresponding Gibbs energy functions from the assessments of binary systems using Muggianu interpolation of binary excess terms [26]. Based on the experimental data of liquid [21], the parameter ${}^{0}L^{\mathrm{liq.}}_{\mathrm{Gd,Mg,Sm}}$ and ${}^{1}L^{\mathrm{liq.}}_{\mathrm{Gd,Mg,Sm}}$ were optimized. Due to a lack of experimental data of hcp and rhombohedral in the ternary

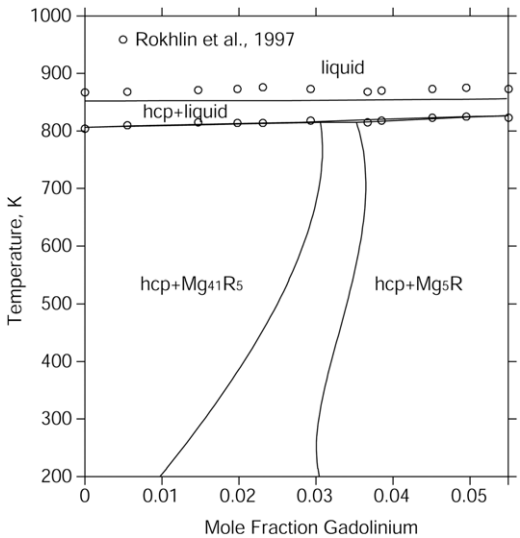

Fig. 7a. Calculated isoplethic section of the Gd–Mg–Sm system at 94.5 at.% Mg using the present thermodynamic description with the experimental data measured by Rokhlin et al. [21].

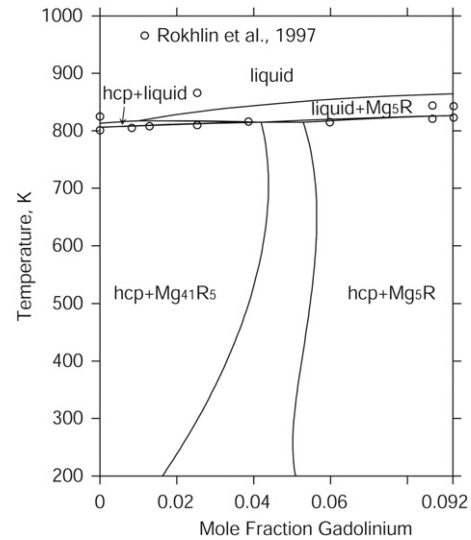

Fig. 7b. Calculated isoplethic section of the Gd–Mg–Sm system at 90.8 at.% Mg using the present thermodynamic description with the experimental data measured by Rokhlin et al. [21].

Table 2
Invariant reactions in the Gd–Mg–Sm system.

Reaction	Type	T (K)	Composition in liquid			T (K)
			x(Gd)	x(Mg)	x(Sm)	Ref. [21]
liq. + Mg_5R → $Mg_{41}R_5$ + hcp	U_1	815	0.0054	0.9724	0.0222	815

system, the ternary interaction parameters ${}^{i}L^{\mathrm{hcp}}_{\mathrm{Gd,Mg,Sm}}$ and ${}^{i}L^{\mathrm{rhomb}}_{\mathrm{Gd,Mg,Sm}}$ are not optimized in the present work. However, in order to assure the liquidus monovariant line between MgR and bcc is smooth, the parameter ${}^{1}L^{\mathrm{bcc}}_{\mathrm{Gd,Mg,Sm}}$ was simply optimized as 10000.0.

The thermodynamic parameters of the compound $Mg_{41}Sm_5$ in the Gd–Mg–Sm system were optimized on the basis of experimental data and thermodynamic parameters of compounds in the binary systems.

The isostructural compounds Mg_5Gd and Mg_5Sm, Mg_3Gd and Mg_3Sm, Mg_2Gd and Mg_2Sm, and MgGd and MgSm were treated as single phases Mg_5R, Mg_3R, Mg_2R and MgR, respectively. Considering the experimental phase relationship [20,21] and the continuity of compounds at room temperature in the ternary Gd–Mg–Sm system, the parameters ${}^{0}L^{\mathrm{MgR}}_{\mathrm{Mg:Gd,Sm}} = {}^{0}L^{\mathrm{MgR}}_{\mathrm{Gd,Sm:Mg}} = +10000.0$, ${}^{0}L^{\mathrm{MgR}}_{\mathrm{Sm:Gd,Mg}} = {}^{0}L^{\mathrm{MgR}}_{\mathrm{Gd,Mg:Sm}} = +15000.0$, ${}^{0}L^{\mathrm{Mg_2R}}_{\mathrm{Mg:Gd,Sm}} = -13000.0$, ${}^{0}L^{\mathrm{Mg_3R}}_{\mathrm{Mg:Gd,Sm}} = -5000.0$ were estimated in the present work.

The optimization was carried out by means of the optimization module PARROT of the thermodynamic software Thermo–Calc [27], which can handle various kinds of experimental data. The program works by minimizing a squared error sum where each of the selected values is given a certain weight. The weight is chosen by our personal judgment and changed by trial and error during the work until most of the selected experimental information is reproduced within the expected uncertainty limits.

5. Results and discussions

The thermodynamic descriptions of the Gd–Mg–Sm system obtained in the present work are shown in Table 1. The calculated temperature and phase compositions at the invariant reactions in the Gd–Mg–Sm system are listed in Table 2.

Fig. 1 is the calculated phase diagrams of the Gd–Mg system using the thermodynamic parameters obtained by Guo et al. [9].

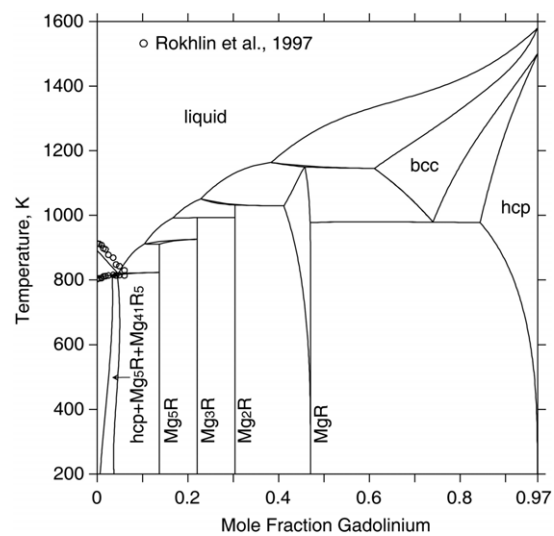

Fig. 7c. Calculated isoplethic section of the Gd–Mg–Sm system at 3 at.% Sm using the present thermodynamic description with the experimental data measured by Rokhlin et al. [21].

Fig. 2 presents the calculated phase diagram of the Mg–Sm system using the present thermodynamic parameters with the experimental data [18,21]. In comparison with the result of Jia et al. [19], the order–disorder transition between MgSm (B2) and bcc (A2) was considered and the solid solubility Sm in hcp(Mg) is closer to the experimental data of Saccone et al. [18].

Fig. 3 presents the calculated phase diagram of the Gd–Sm system using the present thermodynamic parameters. Fig. 4 are the calculated activities of Gd and Sm in hcp at 1173 K with the experimental data [10].

Figs. 5 and 6 present the calculated isothermal sections at 773 and 573 K with the experimental data [20]. As shown in Figs. 5b and 6b, satisfactory agreement was obtained in the Mg-rich corner of the isothermal sections at 773 and 573 K in the Gd–Mg–Sm system.

Figs. 7a–7d shows the calculated vertical sections of the Gd–Mg–Sm system at 94.5 and 90.8 at.% Mg, 3 at.% Sm and $x(\mathrm{Sm})/x(\mathrm{Gd}) = 3$ with the experimental data determined by Rokhlin et al. [21]. The discrepancy between the calculated and experimental data in Figs. 7a–7c was mainly caused because of

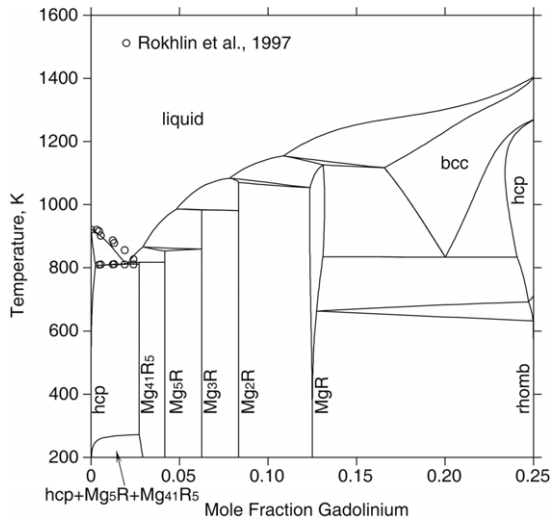

Fig. 7d. Calculated isoplethic section of the Gd–Mg–Sm system at $x(\mathrm{Sm})/x(\mathrm{Gd}) = 3$ using the present thermodynamic description with the experimental data measured by Rokhlin et al. [21].

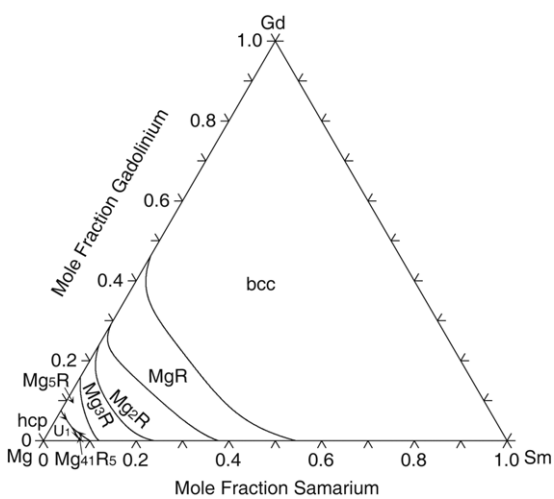

Fig. 8. Calculated projection of the liquidus surface of the Gd–Mg–Sm system using the present thermodynamic description.

the inconsistency between the experimental data determined by Saccone et al. [18] and Rokhlin et al. [21] in the binary Mg–Sm system, which can be seen in the Mg-rich side of Fig. 2.

Fig. 8 is the predicted projection of the liquidus surface of the Gd–Mg–Sm system using the present thermodynamic description.

6. Conclusions

The phase relations and thermodynamic properties of the Mg-rich corner in the Gd–Mg–Sm system were critically optimized on basis of the experimental information available in the literature. A set of thermodynamic parameters describing the Gibbs energies of individual phases in the Gd–Mg–Sm system as functions of composition and temperature was obtained, and the complete ternary phase diagram of the Gd–Mg–Sm system were predicted. With the present optimized parameters, one can now make various thermodynamic calculations of practical interest.

Acknowledgement

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Appendix. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.calphad.2009.12.006.

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